

TrES-3b

R-0021

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260521 · created 2026-07-08T09:57:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461181.8415 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.057
Transit depth (Rp/Rs)^2	3.75 +/- 2.2 [%]
Orbital Inclination (inc)	82.06 +/- 2.77
Airmass coefficient 1 (a1)	3.15 +/- 1.36
Airmass coefficient 2 (a2)	-1.02 +/- 0.49
Scatter in the residuals of the lightcurve fit is	46.52 %
Best Comparison Star	#2 - [564, 397]
Optimal Aperture	1.34
Optimal Annulus	3.25
Transit Duration (day)	0.051 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.194 $\pm$ 0.057 vs 0.16309 (deviation 0.11 σ)
Duration fit vs published	0.051 d vs 0.0567 d (deviation 0.04 σ)
Mid-time O-C	-5.7 min
Depth SNR	0.7
Residual scatter	46.52 %

Light curve

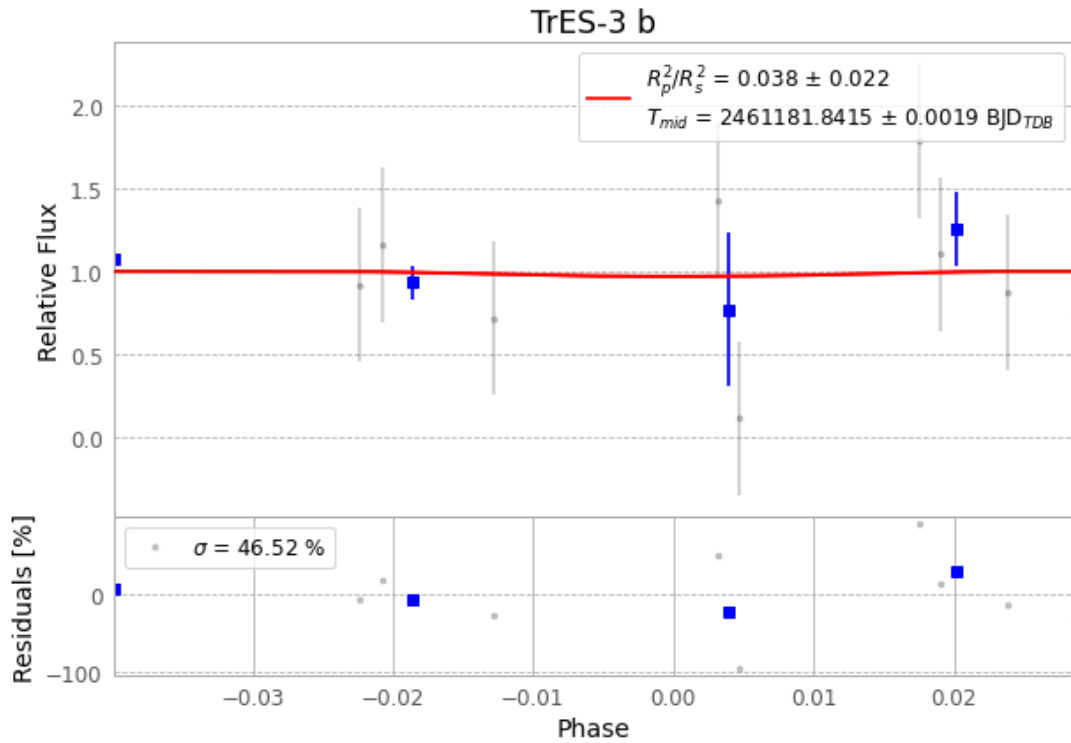


Figure 1 — FinalLightCurve_TrES-3 b_2026-05-21.png (SHA-256 a7423b264d9b8e97...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d6de8c460b9b08d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

68 FITS frames (45 MB), TRES-3260521063015.FITS → TRES-3260521095115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260521.log (7e1d420c7ae5...), FinalLightCurve_TrES-3 b_2026-05-21.png (a7423b264d9b...), FinalLightCurve_TrES-3 b_2026-05-21.csv (d6f869fac060...)

Compute resources

Wall time 136.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.